



# Heterogeneous nucleation of Ni<sub>3</sub>Ti by Mo-enriched particles enhances strength and fracture toughness of maraging steel

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## ABSTRACT

Developing ultra-high strength and high toughness maraging steel requires an effective route of a well-designed heat treatment regimen to control the precipitates. In this study, we regulated the heterogeneous nucleation sites through Mo-enriched particles to increase the nucleation rate of Ni<sub>3</sub>Ti. The Mo-enriched particles as Ni<sub>3</sub>Ti precursors decrease the lattice mismatch between  $\alpha$ -Fe and Ni<sub>3</sub>Ti, reducing crack formation at the interface. We tailored the maraging steel utilizing duplex aging treatment (DAT) to break the conventional precipitation pathway. The DAT comprises a solid solution treatment followed by a two-step aging treatment at 370 °C and 480 °C. The Mo-enriched particles precipitated after solid solution treatment due to the low solubility and diffusion rate at low temperatures. The aging treatment conducted at 370 °C enhances the density of the Mo-enriched particles and inhibits the formation of Ni<sub>3</sub>Ti. The pre-formed precipitates act as heterogeneous nucleation sites for Ni<sub>3</sub>Ti during aging at 480 °C, increasing the nucleation rate and reducing the size. As aging continues, Mo segregates to the interface of Ni<sub>3</sub>Ti and the matrix, inhibiting the growth of Ni<sub>3</sub>Ti and forming Fe<sub>7</sub>Mo<sub>2</sub> precipitates. The DAT steel exhibits an excellent yield strength of 2093 MPa and fracture toughness of 77 MPa•m<sup>1/2</sup> at room temperature. The well-regulated precipitates increase crack nucleation energy, forming a curved primary crack and multiple secondary cracks, which reduce stress concentrations and increase fracture toughness. The DAT strategy provides a new avenue for improving fracture toughness while maintaining high strength in maraging steel.

## 1. Introduction

Maraging steel is a kind of unique ultra-high strength steel, which is strengthened by intermetallic compound precipitations generated on ultra-low carbon Fe–Ni martensite [1–3]. It has excellent mechanical and processing properties and has been widely employed in aerospace, machinery manufacturing, nuclear and military fields [4–6]. The typical application of ultra-strong steel is low-pressure turbine shafts in aero-engine, which are characterized by hollow, thin-walled, high aspect-ratio and complex loads [7,8]. The low-pressure turbine shaft is susceptible to bending deformation and stress concentration during operation [9,10]. As a result, the prepared material must achieve ultra-high strength and excellent fracture toughness [11,12]. However, the fracture toughness of maraging steel is significantly reduced when

the yield strength exceeds 1.9 GPa, which severely limits its engineering applications [13,14]. Therefore, it is imperative to develop maraging steel that demonstrates greater strength and toughness.

The type and morphology of precipitates significantly affect the strengthening mechanism and mechanical properties. Ultra-high strength of maraging steel is achieved by the precipitation of finely dispersed particles [2]. The most commonly formed precipitates include Ni<sub>3</sub>Ti, Ni<sub>3</sub>Mo, laves phase,  $\omega$  phase,  $\sigma$  phase, etc. [15–17]. The morphology of precipitates includes rod, needle, flake, granular, etc. [18,19]. The diversity of alloy compositions and heat treatment processes gives rise to the complexity and variability of precipitates and morphology. Viswanathan et al. [16] demonstrated the formation of the intermetallic phase Ni<sub>3</sub>(Ti,Mo) and Fe<sub>2</sub>Mo by heterogeneous nucleation on dislocations. Niu et al. [14] investigated the Ni–Ti rich clusters

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formed first as precursors to  $\text{Ni}_3\text{Ti}$ , and then the Mo phase wrapped around  $\text{Ni}_3\text{Ti}$  and formed a core-shell structure. Tian et al. [18] indicated that Ni–Ti clusters exhibit a quicker precipitation response than Mo solute, resulting in the formation of a core-shell structure or sphere-like Mo-rich phase. Niu et al. [20] revealed that the formation of  $\text{Ni}_3\text{Ti}$ , Mo-enriched and Cr-rich precipitates is not a separate process and is highly interactive. The conventional precipitation pathway is briefly described as follows: supersaturated solid solution  $\rightarrow$  Ni–Ti cluster  $\rightarrow$   $\text{Ni}_3\text{Ti}$  nucleus  $\rightarrow$   $\text{Ni}_3\text{Ti}$  nucleus and Mo-rich segregation  $\rightarrow$  core-shell structure [18]. The larger  $\text{Ni}_3\text{Ti}$  precipitates produced by conventional precipitation routes in maraging steels can lead to high stress concentrations and microcracks during deformation [20]. Moreover, previous studies could not overcome the trade-off between the strength and toughness of maraging steels [16,21–23].

The high density and low lattice mismatch of nanoscale precipitates facilitate enhanced toughness and ductility of maraging steel [24]. The high-density dispersed nano-precipitates not only pin and obstruct the movement of dislocations, but also serve as a distinctive type of sustainable dislocation source and multiplication [25,26]. Modifying the microstructure and improving the mechanical properties of materials through duplex aging processes is a widely applied strategy for titanium alloys and aluminum alloys [27,28]. Nucleation on pre-existing surfaces decreases the surface energy per unit volume and the interfacial energy of nucleation [29]. Pre-aging increases the nucleation rate and decreases the growth rate of the precipitates, resulting in finely dispersed precipitates [19]. Therefore, a well-designed precipitation path that allows Mo-enriched particles to form preferentially and act as heterogeneous nucleation sites for the later-formed precipitates would ultimately affect the mechanical properties. However, due to the large atomic radius and low diffusion coefficient of Mo, the precipitation pathway mentioned above has not been achieved [30].

In this study, we aimed to regulate the formation sequence of precipitates through the duplex aging treatment (DAT) process. This is implemented by controlling the heterogeneous nucleation sites to increase the nucleation rate and reduce the size of the precipitates. As a result, the maraging steel was processed by DAT routes to obtain different microstructures and mechanical properties. In addition, the in-situ synchrotron X-ray diffraction was used to investigate the variation of dislocation density with strain in DAT steel. To summarize, maraging steel with an excellent combination of ultra-high strength and fracture toughness was obtained, which sheds light on the preparation of high-performance steel.

## 2. Material and methods

### 2.1. Materials preparation

The chemical composition of the maraging steel employed in this study was Fe–18Ni–8.5Co–5Mo–0.7Ti–0.2Al (wt.%). The steel was produced using a vacuum induction melting furnace and cast into ingots with a diameter of 50 mm and a weight of 2 kg. Before melting, the

furnace chamber was evacuated to  $10^{-3}$  Pa and filled with 0.05 MPa of high-purity argon as a shielding gas. The steel ingot was remelted five times to ensure uniform distribution of the alloying elements and subsequently homogenized at 1100 °C for 2 h to eliminate compositional segregation. Specimens for testing and observation were subjected to two different heat treatments. The first heat treatment process was solid solution treatment (SST) at 840 °C for 1 h, followed by single aging treatment (SAT) at 480 °C for 4 h. The second heat treatment process was SST at 840 °C for 1 h, followed by aging at 370 °C for 2 h (DAT-1) and at 480 °C for 4 h (DAT-2), which was denoted by duplex aging treatment (DAT). The heat treatment processes used to produce the SAT and DAT samples in this work are displayed in Fig. 1.

### 2.2. Mechanical property measurement

The dog-bone shaped tensile specimens with a gauge length of 20 mm and a gauge diameter of 3 mm were machined from heat-treated samples. Room temperature tensile tests were conducted on the Instron 5969 testing machine with a laser extensometer at a strain rate of  $1 \times 10^{-3} \text{ s}^{-1}$ . More than three independent tensile tests were performed for each sample. The fracture toughness was assessed by ASTM Standard E1820 using compact tensile C(T) specimens with a thickness of  $B = 4$  mm and a width of  $W = 8$  mm. All specimen surfaces were ground and polished to enable the measurement of crack length using a Leica DM4 long-distance optical microscope. Before the fracture toughness measurement, all the C(T) specimens were fatigue pre-cracked to a crack length with  $0.45W < a < 0.55W$  range using an Instron E300 electrodynamic test instrument at a frequency of 40 Hz with a load ratio  $F_{\min}/F_{\max}$  of 0.1.

The in-situ synchrotron experiments with a gauge size of  $20 \times 4 \times 0.4 \text{ mm}^3$  were performed at Beamline BL22XU in Spring-8, Japan. The wavelength of the high-energy X-ray beam was 0.4131 Å. The beam size was  $0.2 \times 0.2 \text{ mm}^2$ . The sample-to-detector distance was 207.6 mm. A uniaxial tensile machine (Yonekura, Osaka, Japan) with a maximum loading of 2 kN was utilized for the tensile test, and the tensile strain rate was  $1 \times 10^{-3} \text{ s}^{-1}$ . The diffraction profiles (2-D Debye rings) were recorded during the tensile deformation of the samples using a high-speed CCD camera with a dwell time for each profile of 1 s. Then, the recorded images were analyzed using Image J software.

### 2.3. Microstructural characterizations

X-ray diffraction (XRD) characterization was employed on a Bruker D8 X diffractometer. Data collection was performed using  $\text{Cu-K}\alpha$  radiation in the  $2\theta$  range of 30–100° at a scan step interval of 0.02°. The microstructure and fracture topography were observed using the Tescan Clara S8151 scanning electron microscopy (SEM) and energy dispersive spectrometer (EDS). Simultaneously, electron backscatter diffraction (EBSD) mapping was conducted using the Tescan Clara S8151 SEM equipped with a fully automatic Oxford Instruments Aztec 2.0 EBSD system (Channel 5 software). The operating voltages of SEM and EBSD

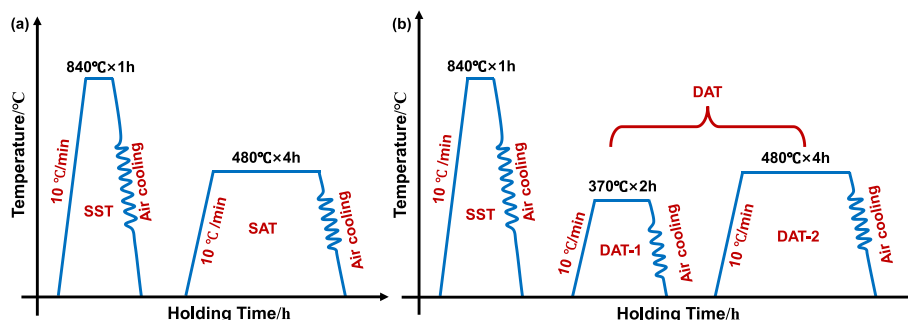


Fig. 1. Illustration of two different heat treatment processes applied to the maraging steel. (a) SST and SAT. (b) SST followed by DAT-1 and DAT-2.

were 20 kV and 10 kV, and the scanning step size of EBSD was 0.05  $\mu\text{m}$ . The EBSD sample was mechanically polished and electropolished in an electrolyte of 95 vol% glacial acetic acid and 5 vol% perchloric acid at  $-25^\circ\text{C}$  and 30 V.

The transmission electron microscope (TEM) observations were conducted in an FEI-Tecna G2 F20 S-TWIN microscope operating at 200 kV. The samples for TEM characterization were prepared by mechanical polishing down to a thickness of 50  $\mu\text{m}$  and then punching them into discs with a diameter of 3 mm. Finally, the discs were electropolished by twin-jet electro-polishing in an electrolyte of 10 vol% perchloric acid and 90 vol% glacial acetic acid performed at  $-30^\circ\text{C}$  and 20 V.

### 3. Results

#### 3.1. Microstructure of maraging steels after different heat treatments

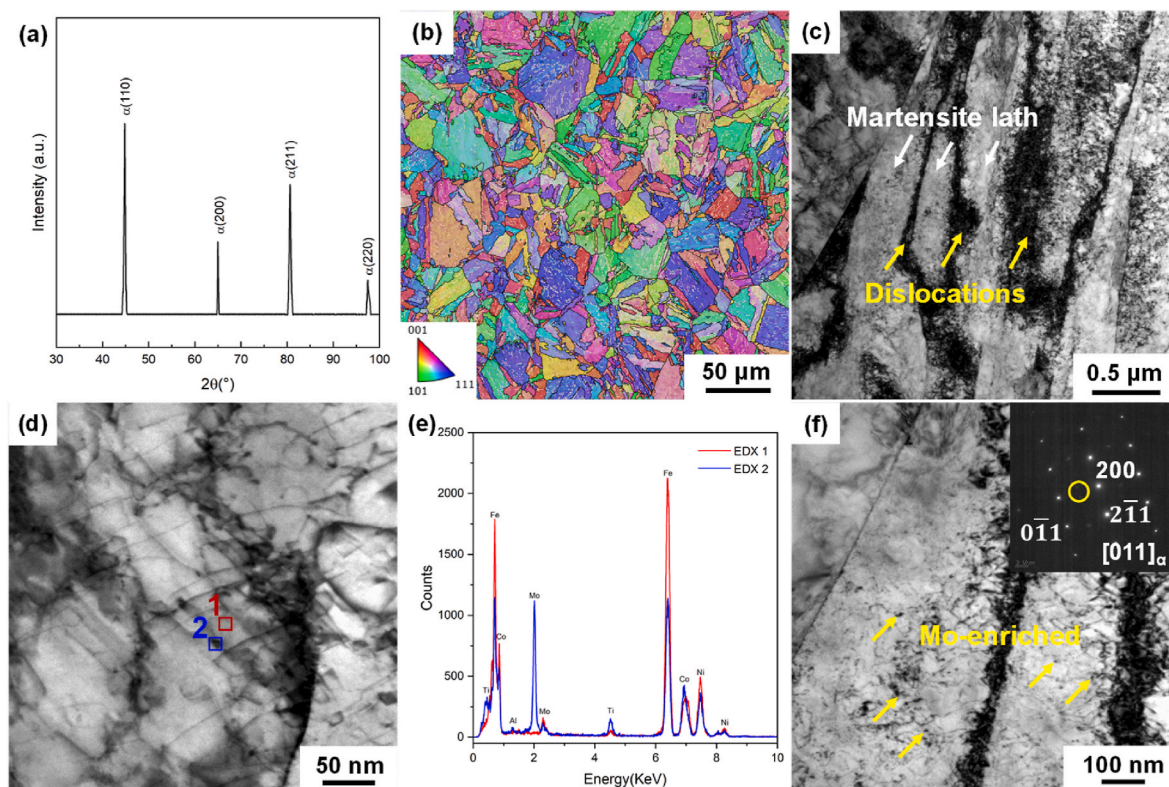
The microstructure of the sample after SST is displayed in Fig. 2. The XRD pattern (Fig. 2a) shows that the SST sample is predominantly composed of martensite. Calculated from the EBSD map in Fig. 2b, the size of prior austenite grains is  $18 \pm 2 \mu\text{m}$  and the width of martensite blocks is  $2.0 \pm 0.8 \mu\text{m}$ . The bright-field TEM (BF-TEM) image reveals that the martensitic laths contain a high density of dislocations, as shown in Fig. 2c. In the high-angle annular dark field (HAADF) image, many entangled dislocations in the martensitic laths and fine precipitates near the dislocations are observed (Fig. 2d). The EDS results acquired from point 1 (matrix) and point 2 (precipitates) demonstrate that the precipitates are enriched in Mo, as depicted in Fig. 2e. The compositions quantified by the EDS points are summarized in Table S1. Extra reflections can be observed in the selected area electron diffraction (SAED) patterns along the  $[011]_\alpha$  direction, demonstrating the existence of precipitates, which are indicated by the yellow arrows in Fig. 2f.

The microstructure of the SAT sample is illustrated in Fig. 3. The BF-TEM image in Fig. 3a reveals a massive dislocation pile-up along with

the martensite lath matrix. Highly dense nanoscale precipitates with acicular morphology are formed within the martensitic lath matrix, as shown in Fig. 3b. The SAED inserted in Fig. 3b displays that the precipitates are  $\eta\text{-Ni}_3\text{Ti}$ , which has a  $\text{D0}_{24}$  ordered structure as indicated in the high-resolution TEM (HRTEM) image and fast Fourier transform (FFT) pattern (Fig. 3c). The orientation relationship between  $\eta\text{-Ni}_3\text{Ti}$  and the matrix is determined as  $\{0001\}_\eta // \{011\}_\alpha$  and  $\langle 11\bar{2}0 \rangle_\eta // \langle 1\bar{1}1 \rangle_\alpha$ , consistent with previous reports [31].

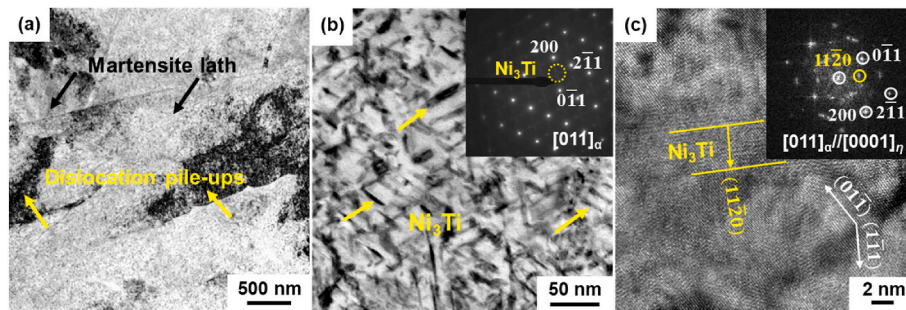
The microstructure of the sample after SST and DAT-1 is indicated in Fig. 4. A high density of tiny spherical precipitates is present in the martensitic matrix of the DAT-1 sample, as displayed in Fig. 4a. In the HAADF image, no other types of precipitates are observed except for the spherical precipitates at the dislocation aggregates, as shown in Fig. 4b. The diameter of these spherical precipitates is approximately 15 nm, calculated using nanometric software and the line intercept method based on HRTEM images (Fig. S1). The EDS maps reveal that the spherical precipitates are enriched in Mo and depleted in Fe and Ni, as illustrated in Fig. 4c. The electron diffraction spots of the precipitates are strong enough to unambiguously identify the pattern compared to Fig. 2f. The diffraction structure indicates that the spherical precipitates are Mo-enriched particles, as depicted in Fig. 4d.

The microstructure of the DAT sample is presented in Fig. 5. However, the microstructure of the DAT sample differs from that of the SAT sample. The high density of clustered dislocations and the presence of reversed austenite at the lath boundaries are observed in the martensitic matrix, as depicted in Fig. 5a. Fig. 5b and the inserted SAED demonstrate the presence of both  $\text{Ni}_3\text{Ti}$  and  $\omega$  phase. According to the dark-field TEM (Fig. 5c), part of the  $\text{Ni}_3\text{Ti}$  formation is individual in the matrix, while the other part is near the  $\omega$  phase. It is verified in HRTEM that  $\text{Ni}_3\text{Ti}$  precipitates at the interface between the matrix and  $\omega$  phase, as shown in Fig. 5d. The FFT patterns (Fig. 5e and f) indicate that the two precipitates correspond to the  $\eta\text{-Ni}_3\text{Ti}$  and  $\omega\text{-Fe}_7\text{Mo}_2$  phases, respectively [2,20]. The orientation relationship between  $\omega\text{-Fe}_7\text{Mo}_2$  and the matrix is

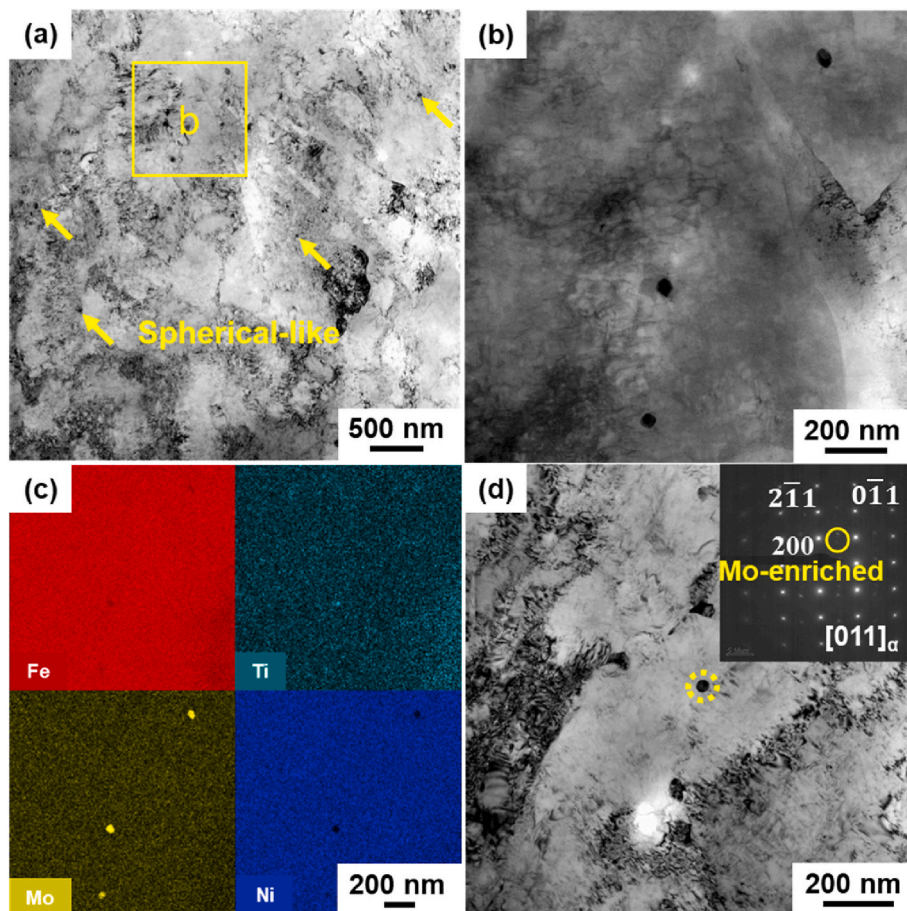


**Fig. 2.** Microstructure of SST sample. (a) The XRD pattern. (b) EBSD band contrast + inverse pole figure + grain boundary map. (c) BF-TEM image. (d) HAADF image shows the precipitates and dislocations. (e) The EDS points of the region are denoted in (d). (f) BF-TEM image and SAED pattern.





**Fig. 3.** Microstructure of the SAT sample. (a) BF-TEM image of dislocations. (b) The enlarged TEM image shows the Ni<sub>3</sub>Ti precipitates and SAED pattern. (c) HRTEM image of the Ni<sub>3</sub>Ti precipitate and FFT pattern.



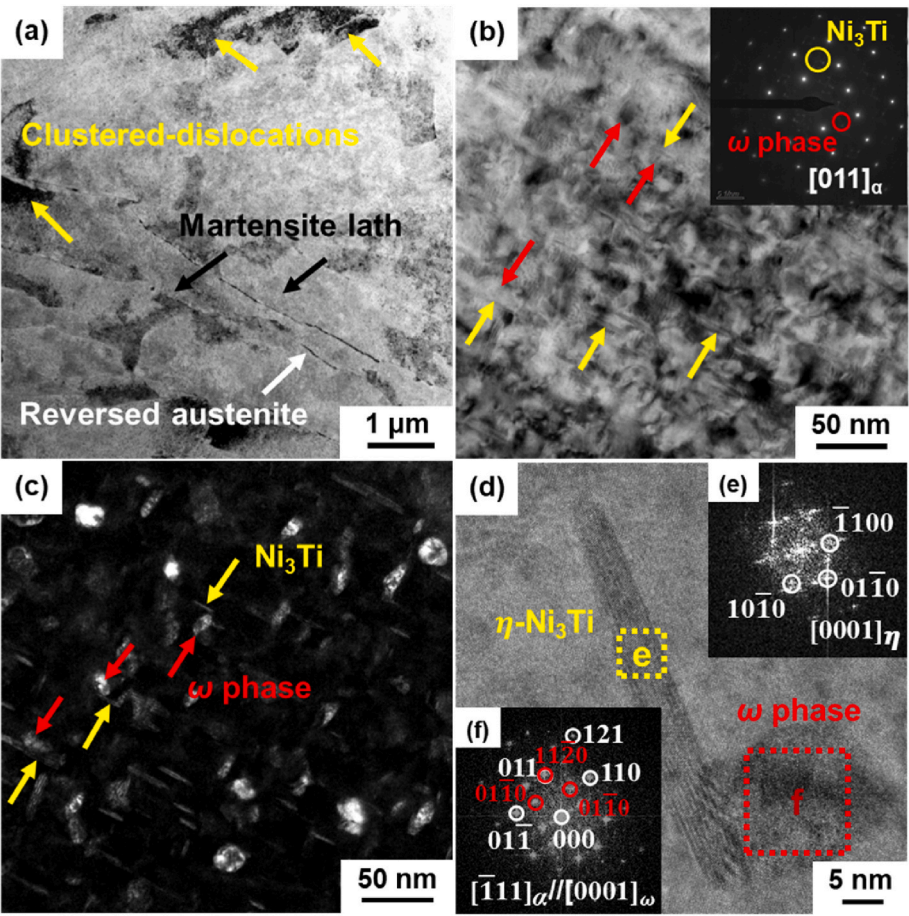
**Fig. 4.** Microstructure of DAT-1 sample. (a) BF-TEM image shows spherical precipitates. (b) HAADF image shows the microstructure of the region denoted in (a). (c) EDS maps of the Fe, Ti, Mo and Ni in (b). (d) BF-TEM image and SAED pattern.

$\{0001\}_{\omega}/\{\bar{1}11\}_{\alpha}$  and  $\langle 11\bar{2}0 \rangle_{\omega}/\langle 011 \rangle_{\alpha}$ , as reported in the literature [32]. The orientation relationship between  $\eta$ -Ni<sub>3</sub>Ti and the matrix is determined as  $\{0001\}_{\eta}/\{011\}_{\alpha}$  and  $\langle 11\bar{2}0 \rangle_{\eta}/\langle \bar{1}11 \rangle_{\alpha}$ , consistent with that in the SAT sample [31].

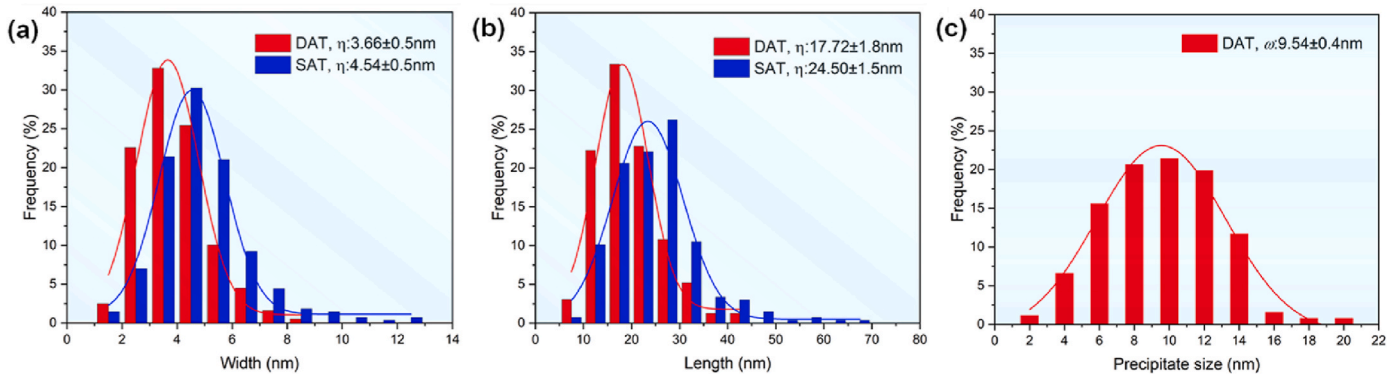
The statistical dimensions of the precipitates following different heat treatments are plotted in Fig. 6, with the results summarized in Table 1. These dimensions are calculated using nanometric software and the line intercept method based on HRTEM images (Fig. S1). According to Fig. 6a, the average width of Ni<sub>3</sub>Ti precipitates is  $4.54 \pm 0.5$  nm in the SAT sample and  $3.66 \pm 0.5$  nm in the DAT sample. Furthermore, the average length of Ni<sub>3</sub>Ti precipitates is  $24.5 \pm 1.5$  nm in the SAT sample and  $17.72 \pm 1.8$  nm in the DAT sample (Fig. 6b). The average width of the Ni<sub>3</sub>Ti precipitates diminishes from 4.54 nm (SAT) to 3.66 nm (DAT).

Similarly, the average length of the Ni<sub>3</sub>Ti precipitates decreases from 24.5 nm (SAT) to 17.72 nm (DAT). Compared to the SAT sample, the width of the Ni<sub>3</sub>Ti precipitate in the DAT sample is reduced by 19.3% and the length by 27.7%. The results illustrate that the size of the Ni<sub>3</sub>Ti precipitate is significantly refined by utilizing DAT. The number density of the Ni<sub>3</sub>Ti precipitates increases from  $2.5 \times 10^{23} \text{ m}^{-3}$  (SAT) to  $5.4 \times 10^{23} \text{ m}^{-3}$  (DAT), and the volume fraction rises from 10% (SAT) to 12% (DAT), as calculated in the supplementary material. Additionally, the volume fraction of the  $\omega$ -Fe<sub>7</sub>Mo<sub>2</sub> in the DAT sample increased dramatically, with a statistical mean diameter of  $9.54 \pm 0.4$  nm (Fig. 6c). On the other hand, due to the uneven distribution and lower content, it is unable to reliably identify Mo-enriched particles in the SAT sample.





**Fig. 5.** Microstructure of the DAT sample. (a) BF-TEM image of dislocations and reversed austenite. (b) BF-TEM image of Ni<sub>3</sub>Ti and ω precipitates. (c) Dark-field TEM image of (b). (d) HRTEM image and the corresponding FFT patterns for the dashed rectangular area.



**Fig. 6.** Statistical results of precipitate size after different heat treatments. (a) The width of η-Ni<sub>3</sub>Ti precipitates in SAT and DAT samples. (b) The length of η-Ni<sub>3</sub>Ti precipitates in SAT and DAT samples. (c) Diameter of ω precipitates in DAT sample.

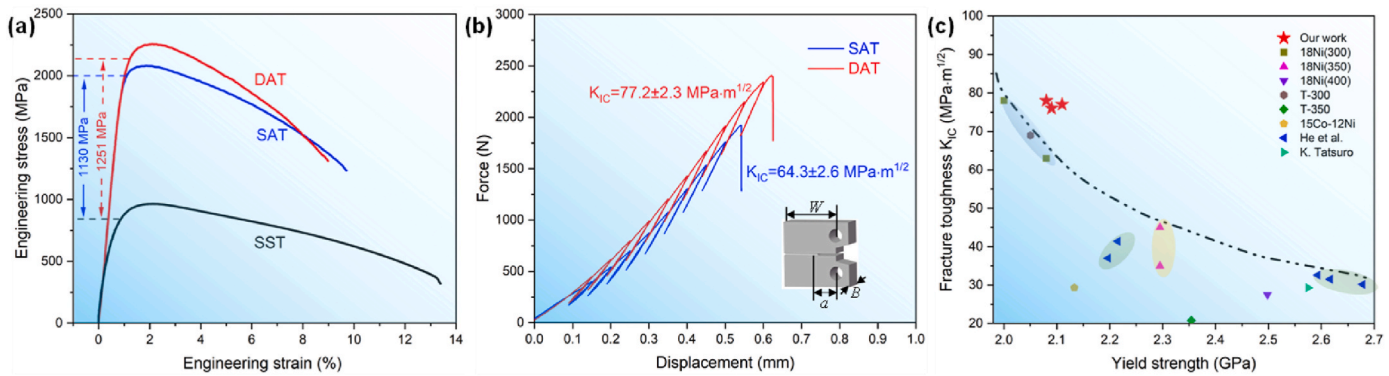
**Table 1**

Average size, number density and volume fraction of precipitates after different heat treatments.

| Samples | Precipitates                      | Width/<br>nm  | Length/<br>nm  | Diameter/<br>nm | Number<br>density/<br>m <sup>-3</sup> | f/<br>% |
|---------|-----------------------------------|---------------|----------------|-----------------|---------------------------------------|---------|
| SAT     | η-Ni <sub>3</sub> Ti              | 4.54 ±<br>0.5 | 24.5 ±<br>1.5  | –               | 2.5 × 10 <sup>23</sup>                | 10      |
| DAT     | η-Ni <sub>3</sub> Ti              | 3.66 ±<br>0.5 | 17.72 ±<br>1.8 | –               | 5.4 × 10 <sup>23</sup>                | 12      |
|         | ω-Fe <sub>7</sub> Mo <sub>2</sub> | –             | –              | 9.54 ± 0.4      | 4.4 × 10 <sup>22</sup>                | 0.2     |

### 3.2. Mechanical properties and plastic deformation behavior

The engineering tensile stress-strain curves of maraging steels after different heat treatments are illustrated in Fig. 7a. The yield strength (YS), ultimate tensile strength (UTS), and elongation (EL) are summarized in Table 2. Following various aging treatments, the YS and UTS are more than twice as high as those of the SST samples. The YS increases from 842 ± 16 MPa in the SST sample to 1972 ± 26 MPa in the SAT sample and further to 2093 ± 17 MPa in the DAT sample, with increases of 1130 MPa and 1251 MPa, respectively. Meanwhile, UTS increases from 964 ± 8 MPa in the SST sample to 2070 ± 23 MPa in the SAT

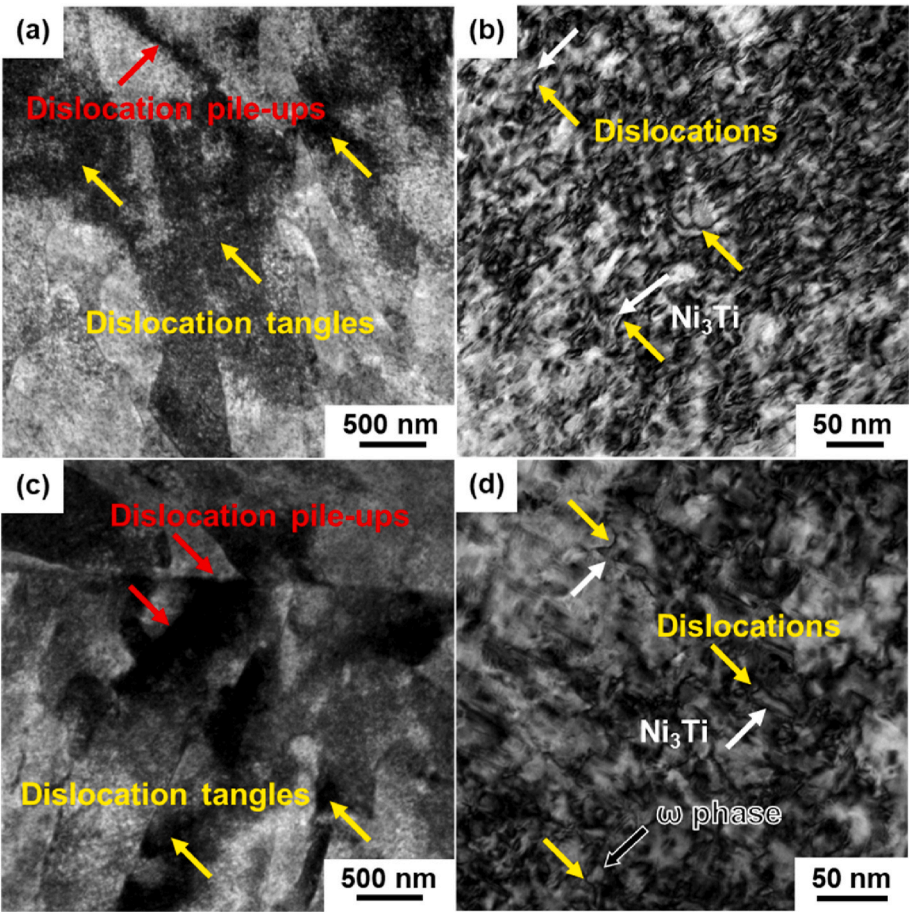


**Fig. 7.** Mechanical properties of maraging steels. (a) Engineering tensile stress-strain curves of the SST, SAT, and DAT samples. (b) Force-displacement curves of the SAT and DAT samples. (c) Ashby map in terms of fracture toughness versus yield strength [33–39].

**Table 2**  
The mechanical properties of maraging steels after different heat treatments.

| Samples | YS/MPa    | UTS/MPa   | EL/%       | $K_{IC}$ /MPa·m <sup>1/2</sup> |
|---------|-----------|-----------|------------|--------------------------------|
| SST     | 842 ± 16  | 964 ± 8   | 13.4 ± 0.5 | –                              |
| SAT     | 1972 ± 26 | 2070 ± 23 | 8.4 ± 1.2  | 64.3 ± 2.6                     |
| DAT     | 2093 ± 17 | 2211 ± 33 | 7.8 ± 0.8  | 77.2 ± 2.3                     |

sample and to 2211 ± 33 MPa in the DAT sample. It is noteworthy that, compared to the SAT sample, the DAT sample's YS and UTS increase by 121 MPa and 141 MPa, respectively. A reduction in the EL was noted, along with an increase in strength. The EL decreases from 13.4% ± 0.5% (SST) to 8.4% ± 1.2% (SAT) and 7.8% ± 0.8% (DAT), respectively. The plane strain fracture toughness ( $K_{IC}$ ) of maraging steels is assessed using C(T) specimens [40]. According to the force-displacement curves, both SAT and DAT samples fracture at the maximum value of the force, as shown in Fig. 7b. It is worth noting that the DAT sample has a large force compared to the SAT sample, suggesting that the DAT sample structure has excellent fracture resistance



**Fig. 8.** BF-TEM images show the microstructure at the gauge sections of tensile fractured specimens. (a) Dislocation distribution in SAT samples. (b) The enlarged TEM image shows the distribution of dislocations and precipitates in the SAT sample. (c) Dislocation distribution in DAT samples. (d) The enlarged TEM image shows the distribution of dislocations and precipitates in the DAT sample.



[41,42]. According to Fig. 7b, the force conforms to the first form in the fracture toughness test criteria, i.e.,  $F_Q$  equals  $F_{max}$  [43]. According to the standard Type I formula (see supplementary material), the  $K_{IC}$  of SAT and DAT samples is  $64.3 \pm 2.6 \text{ MPa}\cdot\text{m}^{1/2}$  and  $77.2 \pm 2.3 \text{ MPa}\cdot\text{m}^{1/2}$ , respectively (Table 2). The results reveal that the DAT sample has excellent fracture toughness compared to the SAT sample. Furthermore, Fig. 7c shows the fracture toughness and YS of the DAT sample compared to other ultra-strong structural materials. It is established that the fracture toughness of structural materials diminishes drastically as YS increases. However, the DAT sample has a high fracture toughness at YS of more than 2 GPa. The DAT sample with high yield strength and fracture toughness is designed based on nanoprecipitate.

The TEM analysis is carried out to reveal the microstructure of maraging steel after deformation, as shown in Fig. 8. The dislocation slip is impeded due to the dense precipitates in martensitic laths, resulting in dislocation tangles (Fig. 8a). Meanwhile, the strong interaction of high-density dislocations at the lath interface induces the formation of dislocation pile-ups, as depicted in Fig. 8a. The  $\text{Ni}_3\text{Ti}$  precipitates can prevent the slip of partial dislocations and leave a bow-shaped dislocation morphology around the precipitates, as displayed in Fig. 8b. The average diameter of  $\text{Ni}_3\text{Ti}$  in the SAT sample, calculated by the iso-volumetric method, is about 9.1 nm. The average diameter exceeds the critical value of  $15b$  ( $b$  is the Burgers vector), suggesting that the

precipitation strengthening mechanism in the SAT sample is an Orowan bypass mechanism [44]. Fig. 8c shows that the dislocations are distributed inside the martensitic lath and pile-ups at the lath interface with a dislocation density of  $13.8 \times 10^{14} \text{ m}^{-2}$  (Fig. S2). The  $\text{Ni}_3\text{Ti}$  and  $\text{Fe}_7\text{Mo}_2$  precipitates can prevent the slip of some dislocations and leave curved dislocation shapes near the precipitates, as illustrated in Fig. 8d. Similarly, the average diameter of  $\text{Ni}_3\text{Ti}$  in the DAT sample is about 7.5 nm, while the diameter of  $\text{Fe}_7\text{Mo}_2$  is about 9.54 nm based on the statistics in Fig. 6c. The diameters of both precipitates exceed the critical value, indicating that the precipitation strengthening mechanism in the DAT samples is also an Orowan bypass mechanism.

In-situ synchrotron X-ray diffraction was conducted to further investigate the relationship between the microstructure and properties of the DAT sample. Fig. 9a shows the relationship between the full width at half maximum (FWHM) and applied stress. For the DAT sample, the FWHM of the planes increases with increasing stress, which is consistent with the report of Hu et al. [45]. Based on the FWHM data (Fig. 9a), the relationship between  $\Delta K$  and  $K\bar{C}^{1/2}$  can be obtained (see supplementary material), as shown in Fig. 9b. Fig. 9c displays the dislocation density calculated for the  $\alpha$  phase in the DAT sample at different tensile strains. It can be seen that the initial dislocation density of the DAT sample is  $\sim 4.47 \times 10^{14} \text{ m}^{-2}$ , which increases significantly as plastic deformation

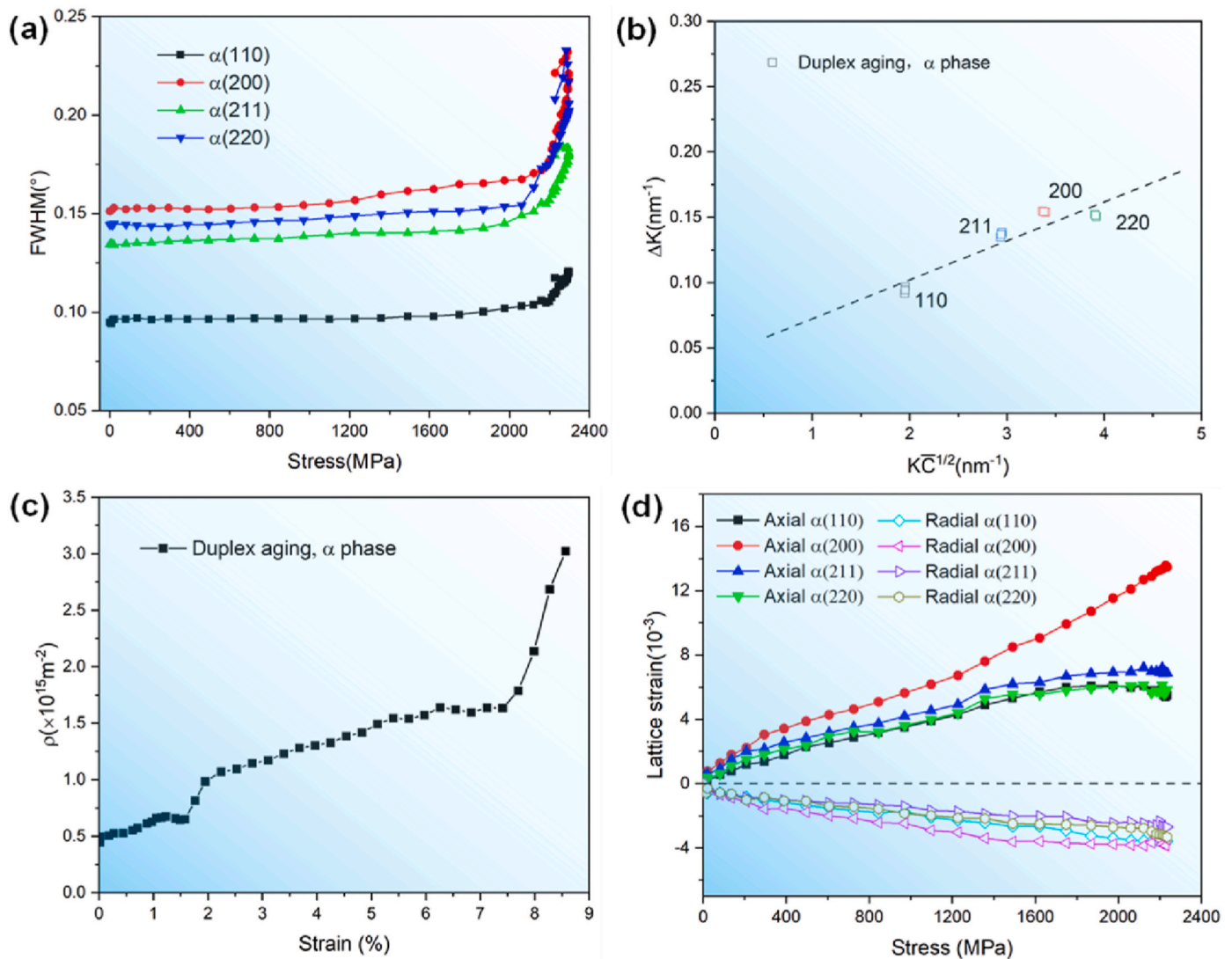


Fig. 9. (a) FWHM of DAT sample versus stress. (b) Modified Williamson-Hall plot. (c) The dislocation density of  $\alpha$  phase as a function of strain. (d) Lattice strain of  $\alpha$  phase for DAT sample in different lattice planes.



proceeds. The evolution of the microstructure of the DAT sample is further investigated to gain insight into the microscopic deformation mechanisms of the DAT sample. The lattice strain of the crystallographic plane can be calculated using Eq. (1) [46]:

$$\varepsilon_{hkl} = \frac{d_{hkl} - d_{hkl}^0}{d_{hkl}^0} \quad (1)$$

where  $\varepsilon_{hkl}$  is the lattice strain,  $d_{hkl}$  and  $d_{hkl}^0$  are determined from the interplanar spacing of the crystallographic plane,  $d_{hkl}^0$  is the interplanar spacing of the crystallographic plane without deformation. Fig. 9d indicates the lattice strain in  $\alpha$  phases of the DAT sample, which can be seen to increase significantly during the loading process. In particular, the lattice strain of the  $\alpha_{(200)}$  plane could reach  $14 \times 10^{-3}$  when the stress is equal to  $\sim 2200$  MPa.

## 4. Discussion

### 4.1. Evolution of microstructure during heat treatment

The size of the precipitates in the DAT sample was smaller than that in the SAT sample, and more importantly, the process of precipitate formation was different in the DAT sample. The higher number density and volume fraction of  $\text{Ni}_3\text{Ti}$  precipitates were attributed to the heterogeneous nucleation sites provided by the Mo-enriched particles. The enrichment of Mo elements was facilitated by the fact that cobalt reduces the solubility of Mo in maraging steel [47] and the low diffusion rate of Mo in  $\alpha\text{-Fe}$  [48]. The formation energy values indicate the preferential segregation of solute atoms, which are 0.133 eV and 0.071 eV for Ni and Mo in  $\alpha\text{-Fe}$ , respectively [49]. As a result, Mo will precipitate from the matrix during the subsequent cooling after SST (Fig. 2f). In the SAT sample, acicular  $\text{Ni}_3\text{Ti}$  precipitates preferentially form near the dislocations and within the martensitic lath during aging at  $480^\circ\text{C}$  (Fig. 3b). This is because the high energy near the dislocations lowers the energy barrier for nucleation [50].

The aging treatment at  $370^\circ\text{C}$  increases the density of the Mo-enriched particles and prevents the formation of  $\text{Ni}_3\text{Ti}$  precipitates (Fig. 4a). The diffusion rate of atoms at  $370^\circ\text{C}$  is significantly low, thereby impeding the occurrence of atomic transitions. The energy provided cannot overcome the phase transformation energy barrier necessary to form  $\text{Ni}_3\text{Ti}$  [51]. The fine precipitates exhibit a tendency to coarsen, thereby forming larger particles in order to minimize the overall interfacial energy [52]. Therefore, when the SST sample was aged at  $370^\circ\text{C}$ , the Mo-enriched particles grew into larger particles (Fig. 4b). Nucleation attached to pre-existing surfaces can decrease the interfacial energy of nucleation [29]. The Mo partitioning increases the chemical driving force and reduces the strain energy for the nucleation of the precipitates [20]. According to the heterogeneous nucleation theory, the Mo-enriched particles are precursors to  $\text{Ni}_3\text{Ti}$  and can act as nucleation sites for  $\text{Ni}_3\text{Ti}$ , lowering the nucleation barrier [53]. During the aging process, the aging temperature provides thermodynamic conditions [19]. Therefore, the Mo-enriched particles formed after DAT-1 serve as heterogeneous nucleation sites for  $\text{Ni}_3\text{Ti}$  precipitates during DAT-2, increasing the nucleation rate and refining the size of  $\text{Ni}_3\text{Ti}$  (Fig. 5b and 6b) [54]. As the aging process at  $480^\circ\text{C}$ , the needle-like  $\text{Ni}_3\text{Ti}$  particles appear not only near the dislocations in the matrix but also at the interface between the Mo-enriched particles and the matrix (Fig. 5c). As the aging process continues, the Mo elements aggregate at the interface of  $\text{Ni}_3\text{Ti}$  and the matrix following aging at  $480^\circ\text{C}$ , limiting the growth of  $\text{Ni}_3\text{Ti}$  due to the low diffusion rate of Mo, the same effect is reported in the literature [20,30].

The precipitation of nanophases within the matrix results in lattice mismatch and induces strain energy at the interfaces. A statistical analysis of the a-axis lattice constants of  $\eta\text{-Ni}_3\text{Ti}$  and the matrix was conducted utilizing HRTEM images. The lattice constant of  $\alpha\text{-Fe}$  was calculated to be  $2.846 \text{ \AA}$ , which was consistent with the lattice

parameter of the  $\alpha\text{-Fe}$  matrix in maraging steel ( $2.864 \text{ \AA}$ ) [55]. The lattice constant for  $\eta\text{-Ni}_3\text{Ti}$  precipitate was calculated as  $a_{\eta\text{-SAT}} = 5.083 \text{ \AA}$  and  $a_{\eta\text{-DAT}} = 5.016 \text{ \AA}$ , respectively [24,56,57]. As previously indicated, the orientation relationship between the matrix and  $\eta\text{-Ni}_3\text{Ti}$  was described as  $\{011\}_\alpha // \{0001\}_\eta$  and  $\langle 1\bar{1}1 \rangle_\alpha // \langle 11\bar{2}0 \rangle_\eta$ . The corresponding planes of  $\{011\}_\alpha // \{0001\}_\eta$  serve as a reference, while the lattice mismatch along the close-packed direction was calculated in accordance with the equation  $\delta = 2(a_\eta - a_\alpha)/(a_\eta + a_\alpha)$  [20], where  $a_\eta$  and  $a_\alpha$  are the lattice constant of  $\eta\text{-Ni}_3\text{Ti}$  along the a-axis and  $\sqrt{3}$ -fold of the lattice constant of  $\alpha\text{-Fe}$ , respectively. The lattice mismatch between  $\alpha\text{-Fe}$  and  $\eta\text{-Ni}_3\text{Ti}$  was estimated to be 3.1% in SAT samples and 1.7% in DAT samples, respectively. Thus, the Mo-enriched particles formed using DAT decrease the lattice mismatch between  $\text{Ni}_3\text{Ti}$  and the matrix, reducing the formation of cracks at the precipitate-matrix interface.

In addition, austenite stabilizing elements diffuse into dislocations and other defect structures during aging, causing micro-segregation in localized areas and lowering the  $A_s$  and  $M_s$  temperatures [58,59]. As a result, more reversed austenite was formed, leading to martensitic transformation during deformation [60].

### 4.2. Strengthening mechanism of maraging steel after different aging treatments

For an in-depth analysis of the deformation mechanism, we performed geometric phase analysis on HRTEM images after different aging treatments [61]. The movement of dislocations is obstructed by precipitates bypassing the particles via the Orowan bypass mechanism, as detailed in Section 3.2. The bending of the dislocations increases line tension and requires greater applied stress to allow the dislocations to glide over the particles [62]. As deformation increases, it produces significant precipitation strengthening and leaves bent dislocation morphology within the matrix, as evidenced in Fig. 10a and d. Dislocations bypass the nanoprecipitates, inducing rapid multiplication of dislocations after deformation [25]. Fig. 10b and e displays the HRTEM images along the  $[110]_\alpha$  axis, where the rectangular and elliptical boxes indicate the positions of  $\text{Ni}_3\text{Ti}$  and  $\omega$  precipitates, respectively. The severe distortion in the vicinity of the precipitates forms high localized strain fields, indicating severe interactions between dislocations and precipitates, as shown in Fig. 10c. Differently, due to the higher number density of  $\text{Ni}_3\text{Ti}$  and  $\omega$  particles in the DAT sample, more localized strain fields are generated during deformation, as depicted in Fig. 10f. At the same time, the reduced size of  $\text{Ni}_3\text{Ti}$  makes the dislocations more susceptible to short-range motions, thus bypassing the precipitate particles [39]. The dislocation and precipitate interactions lead to the accumulation of plastic strain in the vicinity of the precipitates, as illustrated in Fig. 10f. This massive dislocation pile-up in the vicinity of the precipitates is responsible for the high strength of the DAT sample (Fig. 7a).

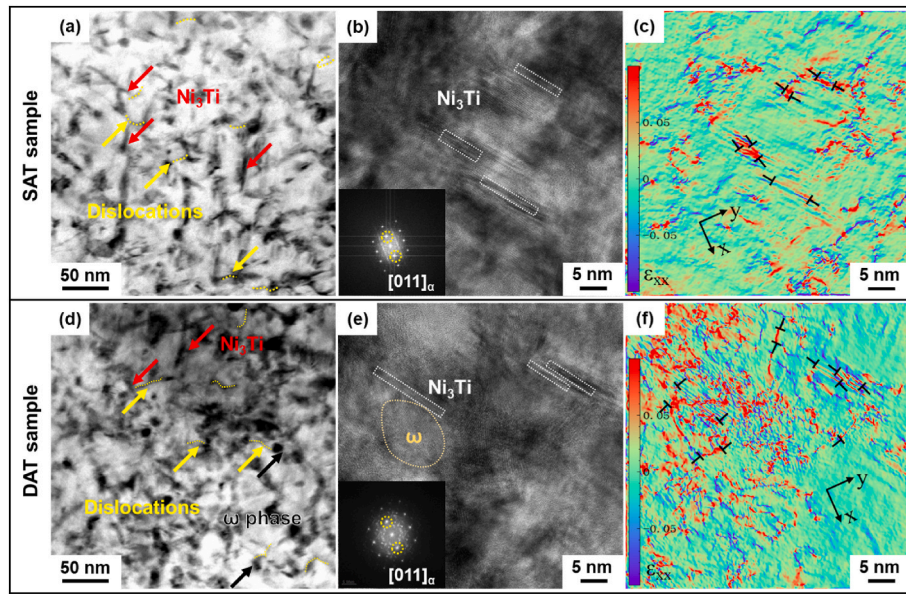
According to the EBSD results (Fig. S2), reverted austenite is less abundant in maraging steels, whereby its effect on strength is neglected [63]. The yield strength of maraging steels mainly stems from four contributions [64]: (1) solid solution hardening  $\sigma_{ss}$ , (2) lath/plate strength  $\sigma_{\text{Mart}}$ , (3) precipitation hardening  $\sigma_p$ , (4) lattice friction stress  $\sigma_0$ . Hence, the yield strength of maraging steels is expressed by the following Eq. (2) [65]:

$$\sigma_Y = \sigma_0 + \sigma_{ss} + (\sigma_{\text{Mart}}^2 + \sigma_p^2)^{\frac{1}{2}} \quad (2)$$

where  $\sigma_0 = 50 \text{ MPa}$  is the lattice friction stress [66],  $\sigma_{ss}$  denotes the strength caused by solid solution, which can be estimated by Fleischer equation [67] as follows:

$$\Delta\sigma_{ss} = \sum (\beta_i^2 x_{i,a})^{\frac{1}{2}} \quad (3)$$

where  $\beta_i$  is the strengthening constant correlated to the modulus and lattice misfit of alloying element  $i$  to Fe. Detailed information on  $\beta_i$  for



**Fig. 10.** Distribution of dislocations after tensile deformation. (a) and (d) BF-TEM images showing the distribution of dislocations and precipitates. (b) and (e) HRTEM images and corresponding FFT patterns taken from the [110]<sub>α</sub> axis. (c) and (f) lattice strain distribution after deformation calculated from lattice images.

the solute atoms of the tested samples can be found in the supplementary material [68,69]. The specific values of  $\beta_i$  are shown in Table S2.  $x_{i,\alpha}$  denotes the atomic fraction of solution element  $i$  in the  $\alpha$ -Fe matrix and can be obtained from the following Eq. (4):

$$x_{i,\alpha} = \frac{x_i - \sum_j x_{i,j} f_j}{1 - \sum_j f_j} \quad (4)$$

where  $x_{i,j}$  represents the equilibrium concentration of the constituent element  $i$  in nanoparticle  $j$ , while  $f_j$  is the volume fraction of particle  $j$ .

The strength of the martensitic matrix is controlled by the variation in grain boundary area and dislocation density. The block is considered the “effective” grain, which governs the grain refinement strengthening of the lath martensite [70]. The Taylor equation accounts for the strengthening contribution of the increase in dislocation density. As proposed by E.I. Galindo-Nava et al. [71],  $\sigma_{Mart}$  can be expressed in terms of Eq. (5):

$$\Delta\sigma_{Mart} = \frac{300}{\sqrt{d_{block}}} + 0.25MGb\sqrt{\rho} \quad (5)$$

where 300 (MPa• $\mu\text{m}^{-1/2}$ ) is the Hall-Petch constant,  $M = 2.73$  denotes the Taylor orientation factor for bcc metals [72],  $G = 71$  GPa is the shear modulus of the Fe–Ni matrix [73],  $b = 0.286$  nm is the magnitude of the Burgers vector,  $d_{block}$  is the average lath thickness of the martensite from EBSD results, and  $\rho$  denotes the dislocation density after aging treatment, respectively.

The precipitates formed during the aging process serve as obstacles to hinder the movement of dislocations and have an outstanding strengthening effect. The mechanisms of precipitation strengthening are related to the type, properties, and size of precipitates. Precipitation strengthening in maraging steels occurs through an Orowan bypass mechanism, according to the results discussed in Section 3.2. Hence, the Orowan-Ashby equation dictates the increase in applied stress for dislocations to bypass fine precipitates [74], as shown in Eq. (6):

$$\Delta\sigma_p = 0.26 \frac{\mu b}{r} f^{\frac{1}{2}} \ln\left(\frac{r}{b}\right) \quad (6)$$

where  $\mu = 0.3$  is the Poisson’s ratio of the matrix.  $r$  and  $f$  are the average radius and volume fraction of the precipitates, respectively.

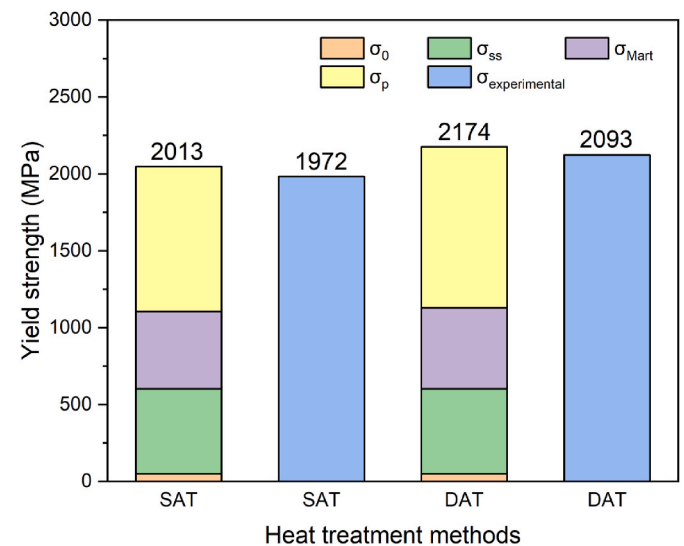
Based on the reasonable assumption that the contributions of the two

types of precipitants are additive, the total precipitation enhancement  $\Delta\sigma_p$  was estimated using the superposition expression [75]:

$$\Delta\sigma_p = \sum_j \left( \Delta\sigma_j^2 \right)^{\frac{1}{2}} \quad (7)$$

where  $\sigma_j$  is the strength increment contributed by precipitate  $j$ .

The experimental results and theoretical predictions of yield strengths for different aging treatments are graphically plotted in Fig. 11. The theoretical calculations of the strengths of the various parts of the SAT and DAT samples are summarized in Table S3. The theoretically calculated values of  $\Delta\sigma_y$  for these samples are 2013 MPa (SAT) and 2174 MPa (DAT), demonstrating only marginal deviations from the experimental values. These results imply that the Orowan bypass mechanism is operative in the SAT and DAT samples.



**Fig. 11.** Estimation results and experimental values of yield strength after different heat treatments.



#### 4.3. Toughening mechanism of maraging steel after different aging treatments

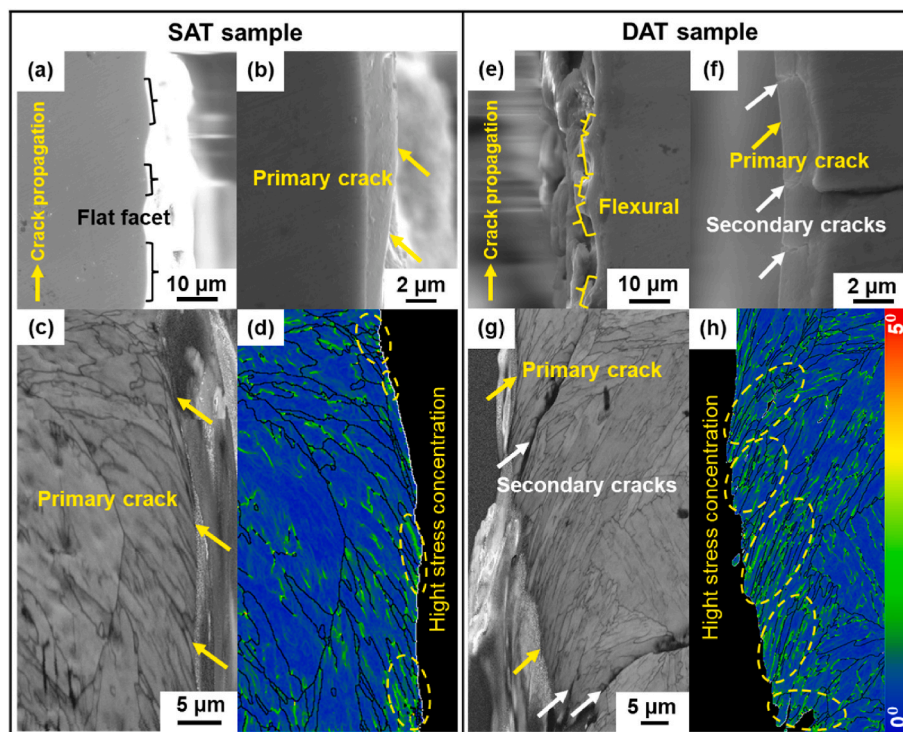
One of the outstanding advantages of the DAT sample was the high fracture toughness combined with considerable strength. The fracture morphology after tensile testing revealed that the DAT samples exhibited a cup-shaped fracture with numerous dimples, which was typical of ductile fracture. The dimples of the SAT samples were large and deep, while those of the DAT samples were small and shallow (Fig. S3). To further elucidate the toughening mechanism, the C(T) fracture toughness samples were analyzed using EBSD and SEM. Fig. 12a shows that the crack propagation path in the SAT sample exhibits an almost flat facet with no secondary cracks (Fig. 12b). However, Fig. 12e displays a more tortuous path of crack extension in the DAT sample, featuring many secondary cracks (Fig. 12f). The primary crack extension is obstructed by fine precipitates, leading to crack deflection and the formation of multiple secondary cracks, thus increasing the crack nucleation energy [76]. The EBSD band contrast image (Fig. 12c) indicates that the crack in the SAT sample crosses the martensitic lath flat and is almost unobstructed. The kernel average misorientation (KAM) image shows fewer and smaller deformation areas, indicating lower toughness (Fig. 12d). However, the crack through the martensitic laths was more curved and generated multiple secondary cracks along the grain or lath boundaries (Fig. 12g). The large and numerous stress concentration regions were observed in the DAT sample, demonstrating greater deformation regions and better toughness [77], as shown in Fig. 12h. A lower lattice mismatch between the precipitates and the matrix, which in turn reduces the potential for cracks at the precipitates-matrix interface, increases the crack nucleation energy [17]. More reversed austenite can be generated owing to the duplex aging process with the aggregation of austenite stabilizing elements and longer aging time (Fig. S2). The transformation of reversed austenite to martensite provided additional transformation-induced plasticity toughening, which can blunt the crack tip and increase crack propagation energy [40,78]. Therefore, these synergistic mechanisms contribute to good toughness in the DAT sample

despite its high yield strength.

Based on the multi-scale characterization results, the microstructure evolution of maraging steel with different aging treatments is schematically illustrated in Fig. 13. The dislocation density of DAT samples decreased slightly compared with the SAT samples (Fig. 13a and d). In the SAT sample, the acicular  $\text{Ni}_3\text{Ti}$  precipitates are preferentially formed near the dislocations and within the martensitic lath (Fig. 13b). This is because the high energy near the dislocations lowers the energy barrier for nucleation. As the aging time increases, some of the Mo elements are segregated at the interface between  $\text{Ni}_3\text{Ti}$  and the matrix (Fig. 13c). However, many tiny spherical Mo-enriched particles form preferentially near the dislocations in the DAT sample (Fig. 13e). The Mo-enriched particles act as heterogeneous nucleation sites for  $\text{Ni}_3\text{Ti}$  particles, increasing the nucleation rate of  $\text{Ni}_3\text{Ti}$  and resulting in fine and uniform particle distribution [54]. As aging proceeds at 480 °C, the acicular  $\text{Ni}_3\text{Ti}$  particles appear not only near the dislocations within the matrix but also at the interface between the Mo-enriched particles and the matrix (Fig. 13f). With the further prolongation of aging, the Mo element aggregates and grows to form the  $\omega$  particles [57]. This DAT increases the density of the precipitates, leading to significant precipitation strengthening. Meanwhile, the refinement of the precipitate decreases the lattice mismatch between the precipitate and the matrix, thereby increasing toughness. In addition, a small amount of reversed austenite is formed due to the aggregation of austenite stabilizing elements and longer aging times (Fig. 13f) [58,60]. The reversed austenite undergoes a TWIP effect during deformation, relieving stress concentration at the crack tip and enhancing the toughness of the DAT sample [79]. Consequently, the DAT breaks the conventional precipitation path and forms finely dispersed precipitates, thereby enhancing the strength and fracture toughness of maraging steels [39].

#### 5. Conclusions

In this study, we investigated the microstructure and mechanical properties of maraging steel after different aging processes. The



**Fig. 12.** Lateral fracture analysis of the C(T) sample after different heat treatments. (a) and (e) SEM images. (b) and (f) Magnified SEM images. (c) and (g) EBSD band contrast images. (d) and (h) KAM images.

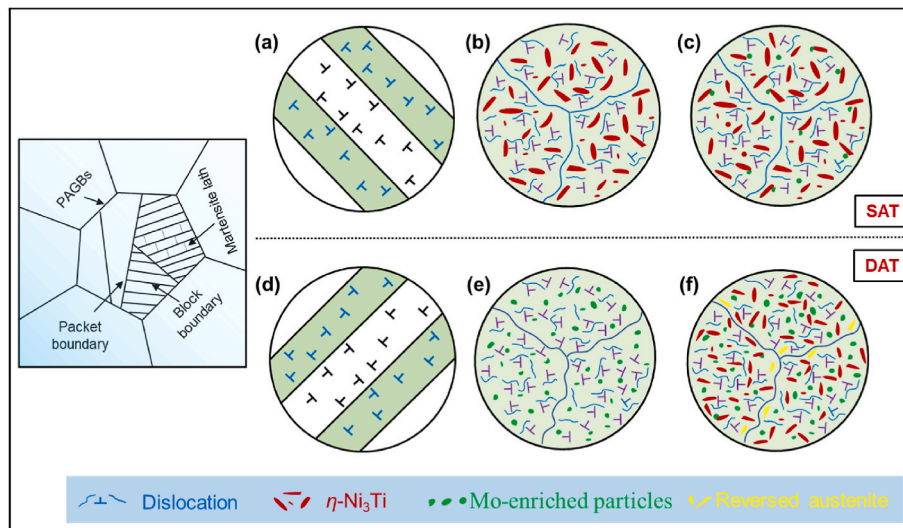


Fig. 13. Schematic diagram of microstructural evolution during different aging treatments.

following conclusions were drawn:

- (1) The solid solution treatment formed tiny Mo-enriched particles, which grow appropriately and increase in density after aging at 370 °C. During aging at 480 °C, Mo-enriched particles act as heterogeneous nucleation sites for  $\eta\text{-Ni}_3\text{Ti}$ , increasing the nucleation rate. The average length of  $\eta\text{-Ni}_3\text{Ti}$  precipitates decreased to 17.72 nm, the number density increased to  $5.4 \times 10^{23} \text{ m}^{-3}$ , and the volume fraction rose to 12% after duplex aging treatment. During further aging, a portion of Mo is segregated to the interface between  $\eta\text{-Ni}_3\text{Ti}$ , inhibiting the growth of  $\eta\text{-Ni}_3\text{Ti}$  and forming the  $\omega$  phase.
- (2) Compared to the solution treatment, the yield strength of maraging steels is significantly increased after different aging treatments. The yield strength of single aging treatment and duplex aging treatment samples is  $1972 \pm 26 \text{ MPa}$  and  $2093 \pm 17 \text{ MPa}$ , while the fracture toughness is  $64.3 \pm 2.6 \text{ MPa}\cdot\text{m}^{1/2}$  and  $77.2 \pm 2.3 \text{ MPa}\cdot\text{m}^{1/2}$ , respectively. The results indicate that the duplex aging treatment sample exhibits an excellent combination of ultra-high strength and fracture toughness.
- (3) The lattice mismatch between  $\alpha\text{-Fe}$  and  $\eta\text{-Ni}_3\text{Ti}$  was estimated to be 1.7% in the duplex aging treatment sample. Theoretical predictions of yield strength imply that the Orowan bypass mechanism is operational in the duplex aging treatment samples. The results of synchrotron radiation experiments demonstrate that the dislocation density of the duplex aging treatment sample continuously increases with the increase in strain. Additionally, the lattice strain in the  $\alpha_{(200)}$  plane undergoes significant changes in the tensile direction.
- (4) The fracture morphology after tensile testing indicates that the duplex aging treatment sample exhibits a typical ductile fracture characterized by a cup-shaped fracture and numerous dimples. The compact tensile test reveals that duplex aging treatment steel exhibits a zig-zag primary crack and multiple secondary cracks, along with large areas of severe high-stress concentration.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmrt.2025.02.139>.

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